

Chapter One: Rotational Motions

Rotational motion: The motion of a rigid body which takes place in such a way that all of its particles move in circles about an axis with a common angular velocity; also the rotation of a body about a fixed point in a space. Circular motion is a rotational motion in which each particle of the rotating body moves in a circular path about an axis. Every point of this rotating body moves in a circle whose center lies on the axis of rotation and every point moves through the same angle during a particular time interval.

Rotational motion is illustrated by

- the fixed speed of rotation of Earth about its axis
- the varying speed of rotation of the fly wheel of a sewing machine
- the rotation of the satellite about a planet
- the motion of an ion in a cyclotron and
- the motion of pendulum.

Rotational variables: (The variables that describe the rotational motion are rotational variables)

A rigid body can simultaneously have two kinds of motion; it can change its position in space that is translational motion and it can change its orientation in space that is rotational motion.

The change of orientation of a rigid body can be regarded as a rotation about some axis. A rigid body of arbitrary shape in pure rotation about the Z axis of a coordinate system is shown in Fig.1, in which z-axis is the axis of rotation. The position of the reference line with respect to the rigid body is arbitrary but it is perpendicular to the rotation axis. To describe the rotational motion of the rotating body the following rotational variables are used:-

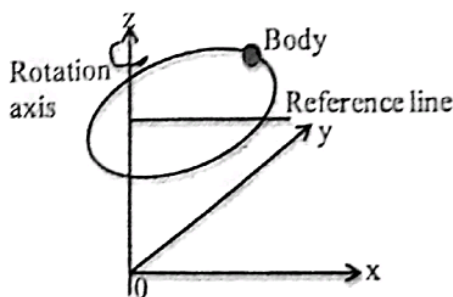


Figure 1

a) **Angular position:** We can describe the motion of the rotating body by specifying the angular position of the reference line relative to the rotation axis. The angular position θ is measured relative to the positive direction of the x-axis and θ is given by $\theta = S/r$ radians, where S is the arc length of a circle of radius r.

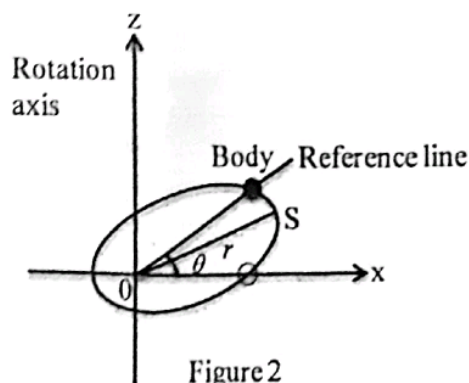
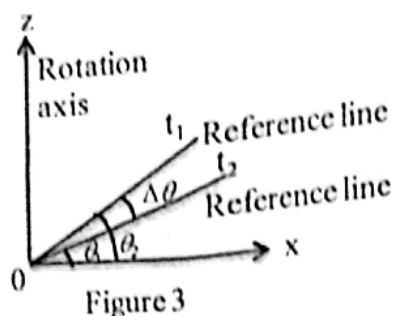


Figure 2

b) **Angular displacement:** If a body rotating about a rotation axis, changes the angular position of the reference line from θ_1 to θ_2 , the body undergoes an angular

displacement $\Delta\theta = \theta_2 - \theta_1$, $\Delta\theta$ is positive for counter clockwise rotations and negative for clockwise rotations.

c) Angular velocity and speed: If a body rotates through an angular displacement $\Delta\theta$ in time interval Δt , its average angular velocity $\bar{\omega}$ is $\bar{\omega} = \frac{\Delta\theta}{\Delta t}$.



The instantaneous angular velocity is $\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$.

d) Angular acceleration: If the angular velocity of a body changes from ω_1 to ω_2 in time interval $\Delta t = t_2 - t_1$, the average angular acceleration $\bar{\alpha}$ of the body is $\bar{\alpha} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t}$.

The instantaneous acceleration is $\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt}$.

Relation between translation and rotational variables and constant acceleration: The translational motion of a rigid body or a particle along a fixed direction (such as x-axis) is the simplest type of motion in which the acceleration a is zero. The next simplest type of motion that corresponds to $a = \text{constant}$ which connect the kinematic variables x , v , a and t in all possible combinations. For the rotational motion of a particle or a rigid body around a fixed axis, the simplest type of motion is that in which the angular acceleration is α zero. The next simplest type of motion that corresponds exactly to linear motion with $\alpha = \text{constant}$ which link the four kinematic variables θ , ω , α and t in all possible combinations. One can derive these angular equations by the methods used to derive the linear equations. Both sets of equations are listed here in the table having $x_0 = 0$ and $\theta_0 = 0$ in these equation for simplicity. Here ω_0 is the angular speed at time $t = 0$.

	Translational motion (Fixed direction)	Rotational motion (Fixed axis)
1.	$v = v_0 + at$	$\omega = \omega_0 + \alpha t$
2.	$x = \frac{v_0 + v}{2} t$	$\theta = \frac{\omega_0 + \omega}{2} t$
3.	$x = v_0 t + \frac{1}{2} at^2$	$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$
4.	$v^2 = v_0^2 + 2ax$	$\omega^2 = \omega_0^2 + 2\alpha\theta$

For the angular quantities, we arbitrarily select one of the two possible directions of motion about the two possible directions of motions about the fixed axis in which θ is increasing. From the equation $\omega = \frac{d\theta}{dt}$ we see that if θ is increasing with time, ω is positive. Similarly, from the equation $\alpha = \frac{d\omega}{dt}$ we see that if ω is increasing with time, α is positive.

Example 1: Prove the equation $\omega = \omega_0 + \alpha t$ for constant angular acceleration.

Solution: Starting from the definition of angular acceleration, $\alpha = \frac{d\omega}{dt}$, we have $\alpha dt = d\omega$ or $\int \alpha dt = \int d\omega$. But α is a constant so that $\alpha \int dt = \int d\omega$. If at $t=0$ we call the angular speed ω_0 , then $\alpha \int_0^t dt = \int_{\omega_0}^{\omega} d\omega$ or $\alpha t = \omega - \omega_0$ or $\omega = \omega_0 + \alpha t$. ✓

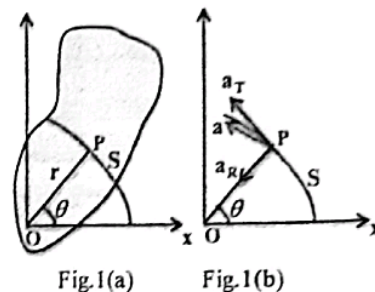
Example 2: A grindstone has a constant angular acceleration α of 3.0 radians/sec². Starting from a horizontal reference line find (a) angular displacement (b) angular speed of stone 2.0 sec later.

Solution: (a) We know $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$
 or $\theta = 0 \times 2.0 + \frac{1}{2} \times 3.0 \times (2.0)^2 = 6.0$ radians (Ans)
 (b) Again $\omega = \omega_0 + \alpha t = 0 + 3.0 \times 2.0 = 6.0$ radians/sec (Ans)

At $t = 0$, $\omega_0 = 0$
 and $\alpha = 3.0$ radians/sec².

★ Relation between linear and angular kinematics for a body in circular motion:

When a rigid body rotates about a fixed axis, every particle in the body moves in a circle. Hence we can describe the motion of such a particle either in linear variables or in angular variables. Consider a particle at P in the rigid body, at a distance r from the axis through O. This particle moves in a circle of radius r as the body rotates as shown in Fig.1 (a). The reference position is ox. The particle moves through a distance S along the arc when the body rotates through an angle θ , such that $S = r\theta$ (1) where θ is in radians.



Differentiating both sides of this equations with respect to the time, taking r is constant, we obtain $\frac{ds}{dt} = \frac{d\theta}{dt} r$ (2)

But ds/dt is the linear speed of the particle at P and $d\theta/dt$ is the angular speed ω of the rotating body so that $v = \omega r$ (3) This is the relation between the magnitudes of the linear velocity and the angular velocity; The linear speed of a particle in circular motion is product ω and r . Differentiating (3) w. r. to t we have $\frac{dv}{dt} = \frac{d\omega}{dt} r$ (4)

But dv/dt is the magnitude of the tangential component of the acceleration of the particle and $d\omega/dt$ is the magnitude of the angular acceleration of the rotating body so that $a_t = \alpha r$ (5)

Hence the magnitude of the tangential component of the linear acceleration of a particle in circular motion is the product of the magnitude of the angular acceleration and the distance r of the particle from the axis of rotation. We have seen that the radial component of the acceleration is v^2/r for a particle moving in a circle. We have $a_r = \frac{v^2}{r} = \omega^2 r$ (6) The resultant acceleration is shown in Fig.1. The above equations describe rotational and linear motions.

Example 3: A grindstone has radius of 0.5m and a constant angular acceleration of 3.0 radians/sec². Starting from a horizontal reference line calculate (a) the linear or tangential speed of a particle on the rim (b) the tangential acceleration of a particle on the rim (c) the centripetal acceleration of a particle on the rim, at the end of 2.0 sec. (d) Are the results same for a particle halfway in the rim, that is, at $r=0.25$ m ?

Solution: We have $\alpha = 3.0$ radians/ sec², $\omega = 6.0$ radians/sec after 2.0 sec and $r = 0.50$ m. Then

(a) $v = \omega r = 6.0 \times 0.5 = 3.0$ m/s.

(b) $a_T = \alpha r = 3.0 \times 0.50 = 1.5$ m/s².

(c) $a_R = v^2/r = \omega^2 r = (6.0)^2 \times 0.50 = 18.0$ m/s².

(d) The angular variables are same for this particle as for a point on the rim. That is once again $\alpha = 3.0$ rad/sec², $\omega = 3.0$ rad/sec. But now $r = 0.25$ m, so that for this particle $v = 1.5$ m/s, $a_T = 0.75$ m/s² and $a_R = 9.0$ m/sec².

Kinetic energy of Rotation and Rotational Inertia: Let us consider a rigid body with angular speed ω about a fixed axis. A particle of mass ' m ' at a distance ' r ' from the axis of rotation moves in a circle of radius ' r ' with an angular speed ω about the axis has a linear speed $v = \omega r$.

Therefore its kinetic energy is $\frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 r^2$. The total kinetic energy of the body is sum of the kinetic energies of its particles. The total kinetic energy K of the rotating body can be written as

$$k = \frac{1}{2}m_1\omega^2 r_1^2 + \frac{1}{2}m_2\omega^2 r_2^2 + \frac{1}{2}m_3\omega^2 r_3^2 \dots \dots \dots (1)$$

$$k = \frac{1}{2}(mr_1^2 + mr_2^2 + mr_3^2 + \dots \dots \dots)\omega^2 \dots \dots \dots (2)$$

$$k = \frac{1}{2}(\sum_{i=1}^n m_i r_i^2)\omega^2 \dots \dots \dots (3)$$

The term $(\sum_{i=1}^n m_i r_i^2)$ is sum over total particles and it is called moment of inertia I of the body. Then $I = \sum_{i=1}^n m_i r_i^2$ is called the rotational of inertia or moment of the body with respect to the particular axis of rotation. The rotational inertia of a body depends on the rotational axis and shape of the body. It's dimension is ML², unit is Kg-m². In terms of rotational inertia, the kinetic energy of the rotating body can be written as $k = \frac{1}{2}I\omega^2 \dots \dots \dots (4)$

Angular momentum of a particle: Consider a particle of mass ' m ' and linear momentum ' P ' at a position ' r ' relative to the origin ' O ' of the inertial reference frame. We define the angular momentum l of the particle with respect to the origin ' O ' to be $\vec{l} = \vec{r} \times \vec{p} \dots \dots \dots (1)$

Angular momentum is a vector. It's magnitude is given by $l = rp \sin\theta \dots \dots \dots (2)$

Where θ is the angle between $\vec{r}$ and $\vec{p}$; its direction is normal to the plane formed by ' $\vec{r}$ ' and ' $\vec{p}$ '. The angular momentum is often called moment of linear momentum. We can derive an important relation between torque and angular momentum. We know that $\vec{F} = \frac{d(m\vec{v})}{dt} = \frac{d\vec{p}}{dt}$ for a particle. Let us take the vector product of $\vec{r}$ with both sides of this expression.

We get $\vec{r} \times \vec{F} = \vec{r} \times \frac{d\vec{p}}{dt}$ (3)

But $\vec{r} \times \vec{F}$ is the torque or moment of force about 'O'. Then we get $\tau = \vec{r} \times \frac{d\vec{p}}{dt}$ (4)

Differentiating both side of equation (1) w. r. to t we get $\frac{d\vec{L}}{dt} = \frac{d(\vec{r} \times \vec{p})}{dt}$ (5)

or $\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}$ (6) But $\frac{d\vec{r}}{dt} = \vec{v}$ is the instantaneous velocity and $\vec{p} = m\vec{v}$.

So that equation (6) becomes $\frac{d\vec{L}}{dt} = \vec{v} \times m\vec{v} + \vec{r} \times \frac{d\vec{p}}{dt}$ (7)

Now $\vec{v} \times m\vec{v} = 0$ because the vector product of two parallel vector is zero. $\therefore \frac{d\vec{L}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt}$... (9)

Which states that the time rate of change of angular momentum of a particle is equal to the torque acting on it. ✓

Moment of Inertia and radius of gyration: According to Newton's first law of motion, a body must continue in its state of rest or uniform motion along a straight line or circle, unless acted upto by an external force. ~~This is the~~ ^{The} inertness or inability of a body to change, by itself, its position of rest or of uniform motion is called inertia, and it is a fundamental property of matter. The mass of a body is thus taken to be a measure of its inertia of motion. (In the case of rotational motion, a body free to rotate about an axis, opposes any change desired to be produced in its state of rest or rotation denotes the rotational inertia or moment of inertia.)

Suppose we have a body of mass 'M', and any rotational axis ZZ' .

The moment of inertia of the particle of mass m_1 at a distance r_1 from the axis is $m_1 r_1^2$. The moment of inertia I , of the whole body about the axis ZZ' , is equal to the sum of $m_1 r_1^2$, $m_2 r_2^2$, $m_3 r_3^2$ etc.

Thus, $I = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 \dots = \sum m r^2$.

or $I = MK^2$, where K is the radius of gyration of the body about the axis ZZ' .

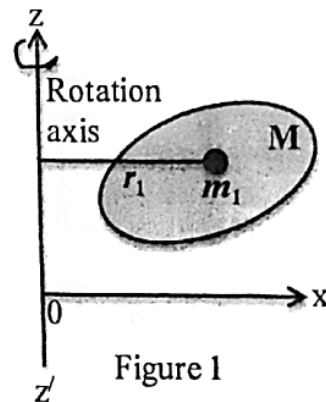


Figure 1

(The effective distance (K) of the particle of a body from its axis of rotation is called its radius of gyration about the axis) and is equal to the root mean square distance of the particles from the axis. The radius of gyration of a body about a given axis of rotation, may be defined as the

distance from the axis at which the whole mass of the body were to be concentrated, the moment of inertia of the body about the given axis of rotation would be same as with its actual distribution of mass.

Torque on a particle If we wish to accelerate the rotation of a body, free to rotate about an axis, we have to apply a couple is called torque that is applied to the body. Considering a rigid body rolling with a uniform angular acceleration $d\omega/dt$, the particle of mass m_1 of the body at a distance r_1 from the axis has a linear acceleration $r_1(d\omega/dt)$. Hence the force acting on the particle is $m_1 r_1 (d\omega/dt)$. The moment of this force about the axis is $m_1 r_1 (d\omega/dt) \cdot r_1 = m_1 r_1^2 (d\omega/dt)$.

Therefore, the moment of this force acting on the entire rigid body about the axis is

$$m_1 r_1^2 (d\omega/dt) + m_2 r_2^2 (d\omega/dt) + m_3 r_3^2 (d\omega/dt) + \dots$$

$$= \frac{d\omega}{dt} (m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + \dots) = I \frac{d\omega}{dt} = I \alpha$$

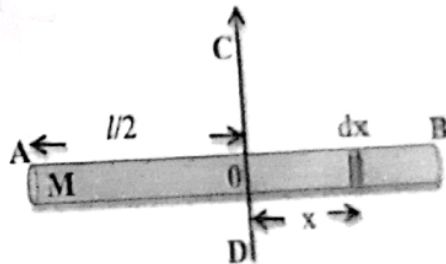
Now, the moment of the force is commonly known as torque, Hence $\tau = I(d\omega/dt) = I\alpha$.

If $(d\omega/dt) = 1$, $\tau = I$.

Thus the moment of inertia of the body about an axis is numerically equal to the torque producing angular acceleration about the same axis.

1. Moment of Inertia of a thin uniform rod about an axis through its center and perpendicular to its length:

Let AB be a thin uniform rod of length 'l' and mass M, free to rotate about an axis CD through its center O and perpendicular to its length. Then its mass per unit length is M/l . Let us consider a small element of length dx of it, at a distance 'x' from 'O'. Its mass is $(M/l)dx$ and its moment of inertia about the axis through O $= (M/l) \cdot dx \cdot x^2$.



The moment of inertia I of the whole rod is obtained by integrating the above expression between the limits $x = -l/2$ and $x = l/2$.

$$\text{Thus } I = 2 \int_0^{l/2} \frac{M}{l} \cdot x^2 \cdot dx = \frac{2M}{l} \left[\frac{x^3}{3} \right]_0^{l/2} = \frac{2M}{l} \left[\frac{(l/2)^3}{3} - 0 \right] \quad \text{or } I = \frac{2M}{l} \cdot \frac{l^3}{24} = \frac{Ml^2}{12}$$

**** Moment of Inertia about an axis passing through one end of the rod and perpendicular to it is**

$$I = \int_0^l \frac{M}{l} \cdot x^2 \cdot dx = \frac{M}{l} \left[\frac{x^3}{3} \right]_0^l = \frac{M}{l} \cdot \frac{l^3}{3} = \frac{Ml^2}{3}$$

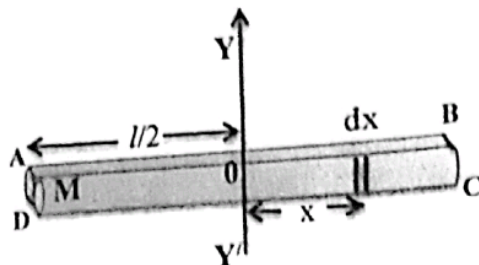
Problem 1: A uniform rod 2 ft long weighing 5 lbs is revolving 60 times a minute about one end. Calculate its moment of Inertia and kinetic energy.

Solution: $I = \frac{Ml^2}{3} = \frac{5 \text{ lbs} \times (2)^2 \text{ ft}^2}{3} = 6.66 \text{ lb. ft}^2$

K.E. $= \frac{1}{2} I \omega^2 = 0.5 I (2\pi n)^2 = 131.31 \text{ ft-pounds}$ where $\omega = 2\pi n$, $n = \frac{60}{60} = 1 \text{ rev/sec}$

- ✱ 2. Moment of Inertia of a Rectangular Lamina about an axis through its center and parallel to its sides.

Solution: Let ABCD be a rectangular lamina and 'l' be its length. Let the axis of rotation YY' pass through its center be parallel to the sides AD and BC. If M be the mass of the rectangle, its mass per unit length will be M/l . Consider a small strip of width dx of mass $(M/l)dx$ at a distance x from the center. Therefore, the moment of inertia of the strip about the axis YY' will be $(M/l)dx \cdot x^2$. The moment of inertia of the whole rectangular lamina about the axis YY' is obtained by integrating within the limits $x=0$ to $x=l/2$ and multiplying by 2.

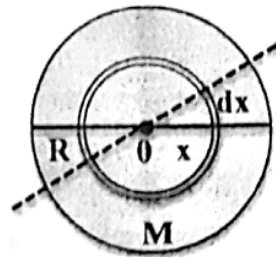


$$I = 2 \int_0^{l/2} \frac{M}{l} \cdot x^2 dx = \frac{2M}{l} \int_0^{l/2} x^2 dx = \frac{2M}{l} \left[\frac{x^3}{3} \right]_0^{l/2} = \frac{2M}{l} \cdot \frac{l^3}{24} = \frac{Ml^2}{12}$$

3. Moment of inertia of an elliptical disc about one of its axes.

- ✱ 4. Moment of inertia of a circular Lamina or Disc about an axis through its center and perpendicular to its plane:

Consider a Disc of radius R and a ring of radius x with thickness dx inside the disc from center O. Its thickness is $2\pi x dx$ and mass is $(M/\pi R^2) \cdot 2\pi x dx$. Hence the moment of inertia of this ring about an axis through 'O' and perpendicular to its plane $= \frac{2Mx \cdot dx}{R^2} \cdot x^2 = \frac{2Mx^3 dx}{R^2}$.



We get the moment of inertia I of the whole disc by integrating the above expression between the limits $x=0$ to $x=R$.

Therefore, the moment of inertia, $I = \int_0^R \frac{2M}{R^2} \cdot x^3 \cdot dx = \frac{2M}{R^2} \left[\frac{x^4}{4} \right]_0^R = \frac{2M}{R^2} \cdot \frac{R^4}{4} = \frac{MR^2}{2}$.

Hence, $I = \frac{MR^2}{2}$.

5. Moment of inertia of a Hollow cylinder about its own axis.

6. Moment of inertia of a spherical shell about its diameter. $I = \frac{2}{3}MR^2$.

7. Moment of Inertia of a solid sphere about its diameter. $I = \frac{2}{5}MR^2$. [D.S. Mathur-Elements of Properties of Matter]

The combined translation and rotation of a rigid body:

We consider a body rotating about an axis and also moving in translational. For example consider a cylinder rolling along a plane surface as shown in Fig.1. If the body rotates with angular velocity ω and moment of inertia I_p , then kinetic energy for rotation can be written as $K = \frac{1}{2}I_p\omega^2$ (1)

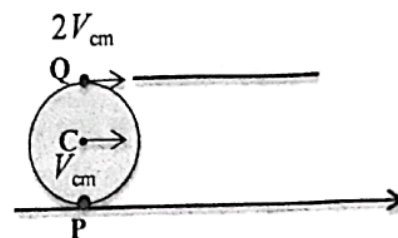


Figure 1

Let us now use the parallel axis theorem, which tells us that

$$I_p = I_{cm} + MR^2 \quad \text{..... (2)}$$

Where I_{cm} is the rotational moment of inertia of the cylinder of mass 'M' and radius 'R' about a parallel axis through the center of mass. Now eqn.1 becomes

$$K = \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}MR^2\omega^2 \quad \text{..... (3)}$$

The quantity $R\omega$ is the speed with which the center of mass of the cylinder is moving with respect to the fixed point 'P'. Let $R\omega = V_{cm}$, the eqn.3 becomes

$$K = \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}MV_{cm}^2 \quad \text{..... (4)}$$

rotational kinetic energy translational kinetic energy

The combined effects of translation of center of mass and rotation about an axis through the center of mass are equivalent to a pure rotation with the same angular speed about an axis through the point of contact of a rolling body.

* Consider a solid cylinder of mass 'M' and radius 'R' rolling down an inclined plane without slipping. Find the speed of its center of mass when the cylinder reaches the bottom.

Conservation of angular momentum:

We know that the time rate of change of the total angular momentum of a system of particles about a fixed point in an inertial reference frame is equal to the sum of the external torques acting on the system i.e. $\tau_{ext} = \frac{dL}{dt}$ (1)

Suppose $\tau_{ext} = 0$, then $\frac{dL}{dt} = 0$, so that $L = \text{constant}$.

When the resultant external torque acting on a system is zero, the total vector angular momentum of the system remains constant. This is the principle of the conservation of angular momentum.

For a system of 'n' particles, the total angular momentum 'L' about some point is

$$L = L_1 + L_2 + L_3 + \dots + L_n \quad (3)$$

$$\text{We have } L = \text{constant} = L_0 \quad (4)$$

Where L_0 is the constant total angular momentum vector.

The angular momentum of the individual particles may change but their sum L_0 remains constant in the absence of a net external torque.

Chapter Two: Oscillatory Motion

Hooke's Law: The Hooke's law states that the force needed to extend or compress a spring by some distance is proportional to that distance. $F \propto x$ or $F = -kx$.

The proportionality constant k is called stiffness constant of spring. This law also states that within elastic limit the strain (deformation) is proportional to stress.

Oscillation: Oscillation is a mechanical phenomenon that occurs about an equilibrium point and the body passes the point from one direction or from two opposite direction in a time period. If each time period is same for oscillation it is called periodic oscillation such as motion of the earth, motion of the simple pendulum.

Vibration: When oscillation occurs about an equilibrium point and the body passes the equilibrium point from two opposite direction in a time period, it is called vibration. Vibration occurs in the motion of a tuning fork or membrane of loud speaker. The periodic motion of a body is said to be vibratory, when it is reversed in direction after a definite interval of time.

Oscillatory motion can be defined as the repeated motion in which an object repeats the same movement and passes a point again and again. The vibratory motion should always be oscillatory but an oscillatory motion may not always be vibratory.

Simple harmonic motion (If an oscillatory particle moves back and forth about an equilibrium position such oscillator is called simple harmonic oscillator and its motion is called simple harmonic motion) The simple harmonic motion (S.H.M) has the following characteristics- **

- i) The motion is oscillatory and is repeated after equal interval of time.
- ii) The restoring force acting on the body is always proportional to its displacement from equilibrium point on its path called mean position.

iii) Its acceleration is always directed towards its mean position.

The characteristic parameters of S.H.M.:

1) Time period: The time taken by an oscillator to complete one oscillation is called time period of oscillation. Time period $T = (2\pi/\omega)$, where ω is angular velocity of oscillation.

2) Frequency: The number of oscillations in one second is called frequency. It is denoted by 'n' or 'f', where $f = (1/T)$.

3) Amplitude: The maximum displacement of the oscillator from its mean position is called amplitude of motion.

4) Phase: The phase of the vibrating particle or oscillator at any instant determines the state of displacement and motion of the particle at that instant.

Differential equation of simple harmonic oscillator:

Suppose the force F is applied on a body of mass ' m '. The displacement of the body is x .

Then from Hooke's law $F \propto -x$. or $F = -kx$ (i)

From Newton's 2nd law of motion $F = ma = m \frac{dv}{dt} = m \frac{d}{dt} \frac{dx}{dt} = m \frac{d^2x}{dt^2}$ (ii)

Equation (i) and (ii) implies that $m \frac{d^2x}{dt^2} = -kx$ or $m \frac{d^2x}{dt^2} + kx = 0$ or $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$

Where $\sqrt{\frac{k}{m}} = \omega \therefore \omega^2 = \frac{k}{m}$ $\therefore \frac{d^2x}{dt^2} + \omega^2x = 0$ (iii)

This is the equation of motion of simple harmonic oscillation. **

Multiplying both sides of equation(ii) by $2 \frac{dx}{dt}$ we obtain

$$2 \frac{dx}{dt} \frac{d^2x}{dt^2} + \omega^2 x \cdot 2 \frac{dx}{dt} = 0 \quad \text{or} \quad \frac{d}{dt} \left(\frac{dx}{dt} \right)^2 + \omega^2 \frac{d}{dt} (x^2) = 0$$

Integrating both sides w.r. to t we get $\left(\frac{dx}{dt} \right)^2 + \omega^2 x^2 = c$ (iv)

Here C is an integrating constant and whose value is determined from initial condition/state of motion. If $x=A$, maximum displacement that is constant, then $\frac{dx}{dt} = 0$, $\therefore C = \omega^2 A^2$ (v)

putting the value of C into (iv) we get $\left(\frac{dx}{dt} \right)^2 + \omega^2 x^2 = \omega^2 A^2$

$$\text{or} \quad \left(\frac{dx}{dt} \right)^2 = \omega^2 A^2 - \omega^2 x^2 \quad \text{or} \quad \left(\frac{dx}{dt} \right)^2 = \omega^2 (A^2 - x^2) \quad \text{or} \quad \frac{dx}{dt} = \pm \omega \sqrt{A^2 - x^2}$$

Considering the positive value $\frac{dx}{dt} = \omega\sqrt{A^2 - x^2}$ or $\frac{dx}{\sqrt{A^2 - x^2}} = \omega dt$ (vi)

Integrating equation(vi) we get $\sin^{-1} \frac{x}{A} = \omega t + \delta$ or $\frac{x}{A} = \sin(\omega t + \delta)$ or $x = A \sin(\omega t + \delta)$

This is the differential equation of S.H.O. where δ is the integrating constant, here phase constant.

Velocity of a particle executing simple harmonic motion: **

We have the expression for displacement $x = A \sin(\omega t + \delta)$.

For initial phase, $\delta = 0$, displacement $x = A \sin \omega t$ or $\frac{x}{A} = \sin \omega t$ (i)

Differentiating $x = A \sin(\omega t + \delta)$ w.r. to t , we get $v = \frac{dx}{dt} = A\omega \cos \omega t$

$$\text{or } v = A\omega\sqrt{1 - \sin^2 \omega t} = A\omega\sqrt{1 - \frac{x^2}{A^2}} = A\frac{2\pi}{T}\sqrt{1 - \frac{x^2}{A^2}} = \frac{A.2\pi}{A.T}\sqrt{A^2 - x^2} = \frac{2\pi}{T}\sqrt{A^2 - x^2}$$

$$= \omega\sqrt{A^2 - x^2} \quad \text{At } x=0, v = \frac{2\pi}{T} A = \omega A, \text{ Velocity is maximum at mean position.}$$

$$\text{At } x=A, v = \frac{2\pi}{T} \cdot 0 = 0, \text{ Velocity is minimum at extreme position.}$$

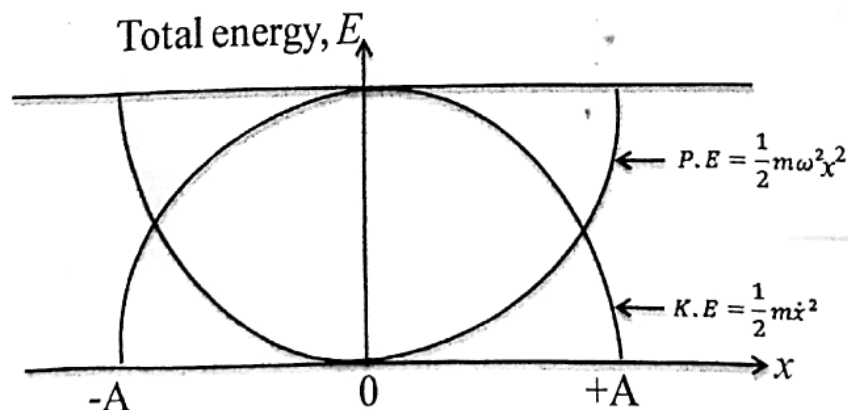
Energy of a particle executing simple harmonic motion: **

A particle executing simple harmonic motion is subject to the action of a restoring force, so its energy can be both kinetic and potential. If no energy is dissipated, it will follow the law of conservation of energy. From the equation of motion, the displacement

$$x = A \sin(\omega t + \delta) \quad \text{..... (i)}$$

$$\text{velocity } v = \dot{x} = \omega A \cos(\omega t + \delta) = \omega\sqrt{(A^2 - x^2)} \quad \text{..... (ii)}$$

$$\text{and acceleration } a = \ddot{x} = -A\omega^2 \sin(\omega t + \delta) = -\omega^2 x \quad \text{..... (iii)}$$



The kinetic energy of a particle of mass 'm' executing simple harmonic motion is $K.E = \frac{1}{2}mv^2$

$$\text{or } K.E. = \frac{1}{2}m(\omega\sqrt{A^2 - x^2})^2 = \frac{1}{2}m\omega^2(A^2 - x^2) \dots\dots\dots (iv)$$

For the extreme position, at $x=A$, the kinetic energy $K.E = \frac{1}{2}m\omega^2 \cdot 0 = 0$

For the mean position, at $x=0$, the kinetic energy $K.E = \frac{1}{2}m\omega^2 \cdot A^2$ that is maximum.

The potential energy of a particle of mass 'm' executing simple harmonic motion is calculated from the total amount of work done for the displacement 'y' is given by

$$P.E. = \int_0^x \text{Force} \times \text{displacement} = \int_0^x m\omega^2 x dx$$

$$\text{or } P.E = \int_0^x m\omega^2 x dx = m\omega^2 \frac{x^2}{2} \dots\dots\dots (v)$$

$$\frac{1}{2}m\omega^2 A^2 - \frac{1}{2}m\omega^2 x^2$$

Thus potential energy is maximum when $x=A$ that is $P.E. = \frac{1}{2}m\omega^2 A^2$ and $P.E.=0$ (min.) at $x=0$.

At any value of x instantaneously $K.E. + P.E. = \frac{1}{2}m\omega^2(A^2 - x^2) + \frac{m\omega^2 x^2}{2} = \frac{1}{2}m\omega^2 A^2 = \text{constant}$.

At any instant, summation of kinetic and potential energy is constant, that proves the principle of conservation of energy.

Average kinetic and potential energy:

The instantaneous value of potential energy is $P.E. = \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t + \delta) \dots\dots\dots (vi)$

The average P.E. for one complete time period is

$$P.E. = \frac{1}{T} \int_0^T \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t + \delta) dt = \frac{m\omega^2 A^2}{4T} \int_0^T 2 \sin^2(\omega_0 t + \delta) dt$$

$$\frac{m\omega^2 A^2}{4T} \int_0^T 1 - \cos 2(\omega t + \delta) dt = \frac{m\omega^2 A^2}{4T} [t]_0^T = \frac{1}{4}m\omega^2 A^2 \dots\dots\dots (vii)$$

Now instantaneous kinetic energy is $K.E. = \frac{1}{2}m\dot{x}^2 = \frac{1}{2}m\omega^2 A^2 \cos^2(\omega t + \delta) \dots\dots\dots (viii)$

The average K.E. for one complete time period is $K.E. = \frac{1}{T} \int_0^T \frac{1}{2}m\omega^2 A^2 \cos^2(\omega t + \delta) dt$

$$\text{or } K.E. = \frac{m\omega^2 A^2}{4T} \int_0^T 2 \cos^2(\omega t + \delta) dt \text{ or } K.E. = \frac{m\omega^2 A^2}{4T} \int_0^T 1 + \cos 2(\omega t + \delta) dt$$

$$\text{or } K.E. = \frac{m\omega^2 A^2}{4T} \int_0^T [t] dt \quad \text{or} \quad K.E. = \frac{m\omega^2 A^2}{4} \dots\dots\dots (ix)$$

The total average energy per one complete time period is $P.E. + K.E. = \frac{m\omega^2 A^2}{4} + \frac{m\omega^2 A^2}{4} = \frac{1}{2} m\omega^2 A^2$.
Thus the average K.E. of a harmonic oscillator is equal to the average P.E. and each is equal to one half of the total energy.

Problem 1: A particle of mass 1 g moves in a potential energy given by $U = U_0 + 6x + x^2$. Find (i) the equilibrium position (ii) force constant (ii) and frequency of oscillation.

Solution: The force is given by $F = -\frac{dU}{dx} = -6 - 2x \dots\dots\dots (1)$

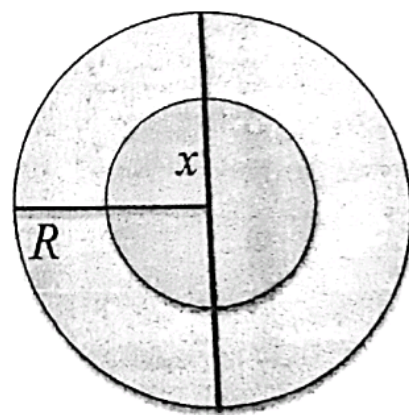
At the equilibrium position force $F=0$, Hence $-6-2x=0$ or $x=-3$ is the equilibrium position.

ii) The equation of motion is $m \frac{d^2x}{dt^2} = -6 - 2x = -2(3 + x) \dots\dots\dots (2)$

Changing the variable by putting $x+3 = y$ or $\frac{dx}{dt} = \frac{dy}{dt}$ or $\frac{d^2x}{dt^2} = \frac{d^2y}{dt^2}$ then equation of motion becomes $m \frac{d^2y}{dt^2} = -2y$. From the relation $F = -kx$, the force constant is 2 dyne/cm.

iii) The frequency of oscillation $\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{2}{1}} = 0.225 \text{ Hz}$.

Problem 2: Assuming the earth as a homogeneous sphere of radius R and a straight hole were drilled through the center of it, show that a ball dropped into the hole will execute S.H.M. and find the frequency of oscillation of the ball.



Solution: Let the ball be at a distance 'x' from the center of the earth where the force of attraction is executed by the inner sphere of volume $\frac{4}{3}\pi x^3$. The mass of the sphere is $\frac{4}{3}\pi x^3 \rho$, where ρ is uniform density. The force acting on the ball is $W = F = \frac{4}{3}\pi x^3 \rho \cdot m \cdot \frac{G}{x^2} = \frac{4}{3}\pi \rho m G x$, where 'm' is the mass of the ball. Here $F = \text{Constant} \times x$. This force is directed towards the center of the earth. The ball constitutes the simple harmonic motion because the force is proportional to the 'x' and it is directed towards the equilibrium position.

The frequency of oscillation is $\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{4}{3} G \pi \rho}$

Now, if 'R' is the radius of the earth, then

$$\frac{G \cdot \frac{4}{3}\pi R^3 \rho \cdot m}{R^2} = mg \quad \text{or} \quad \frac{4}{3} G \pi \rho = \frac{mg}{Rm} = \frac{g}{R}, \text{ Hence } \nu = \frac{1}{2\pi} \sqrt{\frac{g}{R}}$$

Where $F = -kx$

$$\frac{4}{3} \pi \rho m G x = -kx$$

or $k = \frac{4}{3} \pi \rho m G$

* Combination of Simple harmonic oscillations:

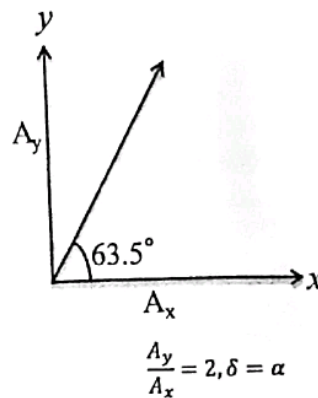
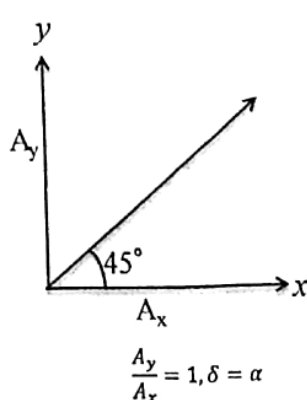
Let us consider two linear simple harmonic motion at right angles are combined. The resultant motion is sum of two independent oscillations. Consider the first case in which the frequencies of the vibrations same such that

$$x = A_x \cos(\omega t + \delta)$$

$$y = A_y \cos(\omega t + \alpha) \dots\dots\dots (i)$$

If the phase constants are same so that $\delta = \alpha$, the resulting motion is a straight line. This can be shown analytically by eliminating 't' from $x = A_x \cos(\omega t + \delta)$ and $y = A_y \cos(\omega t + \alpha)$.

We obtain $\frac{y}{x} = \frac{A_y}{A_x}$ or $y = \left(\frac{A_y}{A_x}\right)x \dots\dots\dots (ii)$ This is the equation of straight line with slope $\frac{A_y}{A_x}$.



Let us show resultant motion for two cases; $\frac{A_y}{A_x} = 1$ and $\frac{A_y}{A_x} = 2$.

In these cases both the x- and y- component displacements reach a maximum value at the same time and reach a minimum at the same time. They are in phase. If the phase constants are different, the resulting motion will not be straight line.

* **Damped harmonic oscillation:** If we assumed that no frictional force act on a oscillator, a simple harmonic oscillator would oscillate indefinitely. Actually the amplitude of oscillation gradually decreases to zero as a result of friction from air or internal force. The motion is said to be damped by friction and is called damped harmonic motion. The equation of motion of damped simple harmonic oscillation is given by the Newton's second law of motion, $F=ma$, in which F is sum of the restoring force, $-kx$ and the damping force $\frac{-b dx}{dt}$. Here b is a positive constant. We obtain $F = ma \dots\dots\dots (1)$

$$\text{or } -kx - b \frac{dx}{dt} = m \frac{d^2x}{dt^2} \text{ or } m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0 \dots (2)$$

If b is small the solution of this differential equation is given by

$$x = Ae^{\frac{-bt}{2m}} \cos(\omega t + \delta) \dots\dots\dots (3) \quad \text{Where } \omega = 2\pi\nu = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2} \dots\dots\dots (4)$$

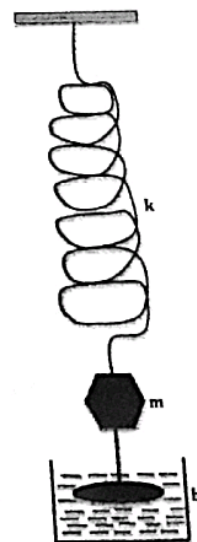


Figure: A damped harmonic oscillator

In figure 2 we plot the displacement x as a function of time ' t ' for oscillatory motion with small damping. We interpret the solution as follows. When friction is present the frequency is smaller and the period is longer. The friction slows down the motion and the amplitude gradually decreases to zero. The amplitude factor is $Ae^{\frac{-bt}{2m}}$, so that $\tau = \frac{2m}{b}$. In

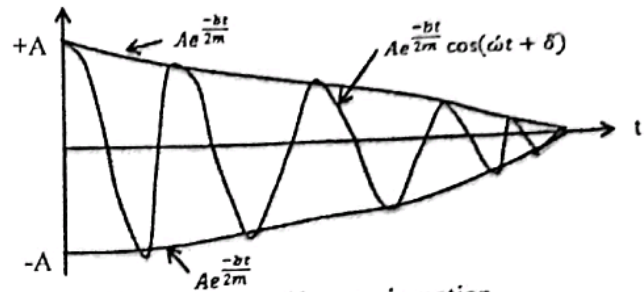


Figure 2: Damped harmonic motion

damped harmonic motion the energy of the oscillator is gradually dissipated by friction and fall to zero in time.

If no frictions were present, b would be equal to zero and ω would equal to $\sqrt{\frac{k}{m}}$ or ω , which is the angular frequency of the undamped motion. If there were no friction present, $b=0$ and the amplitude would have the constant value A as the time went on; the life time would be infinite.

Chapter 3: Surface tension

Molecular force: The interaction between molecules has both attractive and repulsive components, the former being dominant at larger distances and the latter increases very rapidly as the separation decreases normal equilibrium value. Hence there are two types of molecular forces (i) force of adhesion (ii) force of cohesion.

* i) **Force of adhesion:** Adhesion is the force of attraction between molecules of different substances) e.g. gum has a greater adhesive force than water of alcohol.

* ii) **Cohesive force:** Cohesion is the force of attraction between molecules of the same substance) This force depends on the distance between the two molecules and thus rapidly decrease with distance. It is greatest in solids, less in liquids and least in gases. This explains why a solid has a finite shape, a liquid has a definite free surface and a gas has neither. *solid > liquid > gasses.*

Molecular range: The maximum distance upto which the force of cohesion between two molecules can act is called their molecular range. It is actually generally 10^{-7} cm in case of solid. In case of liquids and gas it differ greatly and depends on substances. A sphere drawn around a molecule as center, with a radius equal to its molecular range is called the sphere of influence.

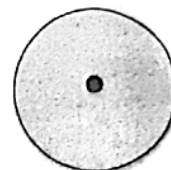


Figure 2: Molecular range

* **Surface tension:** It is a general experience that a liquid in small quantity, free from any external force will always assume the form of a spherical drop, such as rain drop, small quantities of mercury placed on a glass plate. Thus, a liquid always tends to have the least surface area. The tension or pull in the surface of liquid towards center is called surface tension) It is defined as force per unit length of a line drawn in the liquid surface, acting perpendicularly to it at every point and tending to pull the surface apart along the line. λ

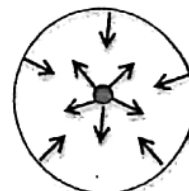
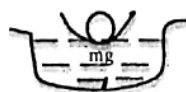


Figure 1: Surface tension Figure 2: Molecular force

Unit and dimension o surface tension: The surface tension may be expressed as force per unit length or energy per unit area. Therefore its unit is Newton/meter of J/m^2 . i.e. $\frac{MLT^{-2}}{L} = MT^{-2}$.

Molecular explanation of surface tension: Let us consider four molecules, A, B, C, and D of a liquid, with their spheres of inclusions drawn around them as shown in Fig.1. The sphere A being well inside the liquid; B is near the free surface of the liquid; C is just near the free surface and D is above the free surface. Since the sphere of influence of A lies wholly inside the liquid, it is attracted equally in all directions by the other

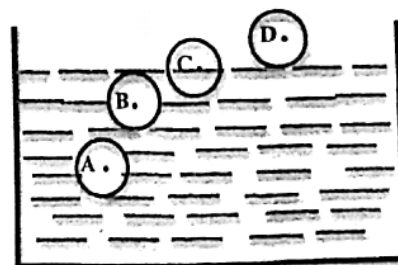


Figure: Molecular Scheme of Surface tension.

molecules lying within its sphere of influence, so that there is no resultant cohesive force on it. On the other hand the sphere of influence of B lies partially outside the liquid, so that the upper half of the sphere contains fewer molecules attracting it upwards and the lower half attracting it downwards, and so there is a resultant downward force acting on B.

The molecule C lies on the surface of liquid, so that full one half of its sphere of influence lies above the surface of the liquid, where as there are liquid molecules in its entire lower half, and thus, the resultant downward force in this case is maximum. This downward or inward force exerted per unit area of a liquid surface is called its internal or intrinsic or cohesive pressure.

In the case of molecule D which is just over the liquid surface, only a part of the sphere of influence lies inside the liquid, so that the downward force on it is very less. Thus we see that all over the surface of the liquid there is a downward pull due to attraction between molecules.

Surface film and surface energy: (If a plane CD be drawn parallel to the free surface layer AB of a liquid at a distance equal to molecular range from it, the layer of the liquid lying in between the free surface and this plane is called the surface film)

Clearly all molecules in the film are being acted upon by a resultant cohesive force in the downward direction. Therefore the potential energy of the molecules in the surface film is greater than that of those inside the liquid. Since, however, the potential energy of a system tends towards a minimum, the film tends to have the least surface area in order that the number of molecules in it may be a minimum. (The potential energy per unit area of the surface film is called its surface energy. Hence to extend the surface area the work done against the cohesive forces among the molecules in the surface is called the surface energy)

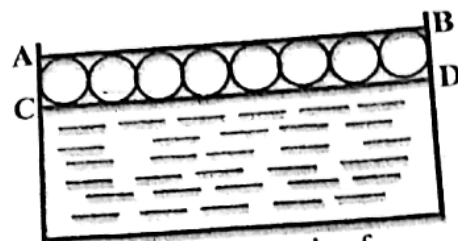


Figure: Schematic of Surface film.

Pressure difference across a liquid surface:

(a) Excess pressure inside a Liquid drop: We know that the molecules near the surface of the drop, experience a resultant pull inwards. Let the excess pressure inside over the pressure outside the drop be 'p'. Then, if 'r' be the radius of the drop and 'T' be its surface tension, we have considering the upward thrust on the upper hemisphere ABCD due to the excess pressure p is equal to $p\pi r^2$. Then the force due to surface tension, acting downwards on it and around its edge is equal to $T.2\pi r$. Since the hemisphere is in equilibrium we have $p\pi r^2 = T.2\pi r$ or $p = \frac{T.2\pi r}{\pi r^2} = \frac{2T}{r}$.

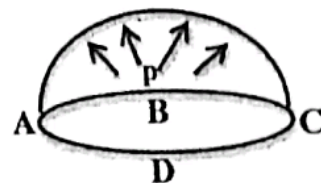


Figure: Schematic of Excess pressure.

Excess pressure inside a Soap bubble: If instead of a liquid drop, we consider a bubble, there are two surfaces to be considered because it is like a spherical shell or a hollow sphere having an inner and outer surface so that the force due to surface tension is $2 \times 2\pi rT$.

In equilibrium we have $p\pi r^2 = 2 \times 2\pi r.T = 4\pi r.T$ or $p = \frac{4\pi rT}{\pi r^2} = \frac{4T}{r}$

Curvature, Pressure and Surface tension:

Consider a small curvilinear element ABCD of a liquid surface film at rest. This film exist only when the inward pressure on it due to surface tension is just balanced by the outward excess pressure on its inner surface. Let us calculate this excess pressure p . Let the length of adjacent sides AB and BC be x and y , Their radii of curvature r and their center of curvature O. Then the surface area of the curvilinear element ABCD = xy , and outward force on it due to excess pressure ' p ' is equal to pxy . Suppose the element is displaced parallel to itself through a small distance δr into the position A'B'C'D'.

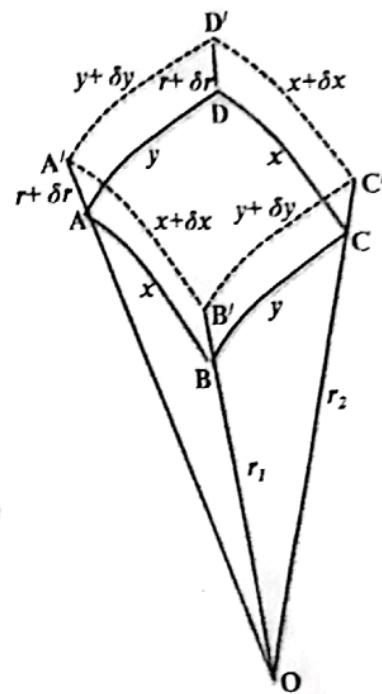


Figure: A curved surface of film

Then the work done during the displacement = $p \cdot xy \cdot \delta r$.

If the length of the sides AB and CD be now $x + \delta x$ and $y + \delta y$, respectively, the new area of the element is obviously $(x + \delta x)(y + \delta y)$. The increase in surface area of the element

$$= (x + \delta x)(y + \delta y) - xy = x \cdot \delta y + y \cdot \delta x$$

The increase in surface energy of the element = $T \cdot (x \cdot \delta y + y \cdot \delta x)$, where T is the surface tension of the liquid. Thus we have $p \cdot xy \cdot \delta r = T \cdot (x \cdot \delta y + y \cdot \delta x)$

$$\text{or } p = T \left(\frac{1}{y} \cdot \frac{\delta y}{\delta r} + \frac{1}{x} \cdot \frac{\delta x}{\delta r} \right) \dots\dots\dots (i)$$

Now, the element being small and A'B' being parallel to AB, so that $\frac{A'B'}{AB} = \frac{BO}{BO}$.

$$\text{or } \frac{x + \delta x}{x} = \frac{r_1 + \delta r}{r_1} \text{ or } \frac{\delta x}{x} = \frac{\delta r}{r_1} \text{ or } \frac{1}{x} \cdot \frac{\delta x}{\delta r} = \frac{1}{r_1} \dots\dots\dots (ii)$$

$$\text{In similar way we have or } \frac{y + \delta y}{y} = \frac{r_2 + \delta r}{r_2} \text{ or } \frac{\delta y}{y} = \frac{\delta r}{r_2} \text{ or } \frac{1}{y} \cdot \frac{\delta y}{\delta r} = \frac{1}{r_2} \dots\dots\dots (iii)$$

$$\text{Now, substituting these values into (i) we have } p = T \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \dots\dots\dots (iv)$$

$$\text{If } r_1 = r_2, \text{ then } p = \frac{2T}{r}.$$

Shape of Liquid meniscus in a capillary tube:

We know that a liquid meniscus in a capillary tube is concave for a liquid like water and convex for a liquid mercury. Let a capillary tube of glass be dipped vertically in a liquid, meeting its surface at P. Then a liquid molecule at P, in contact with the tube, will be attracted outward by the solid molecule due to force of adhesion. The same liquid molecule will be attracted inward by the near molecule of the liquids due to force of cohesion.

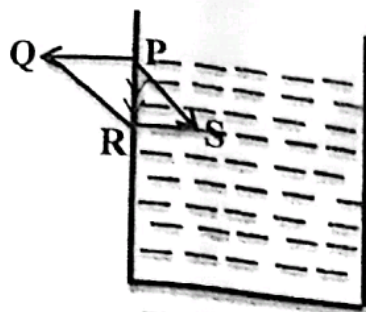


Figure1

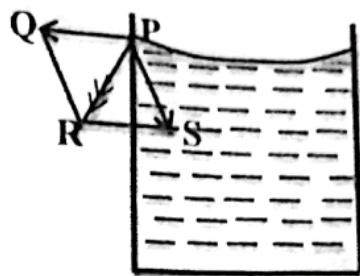


Figure2

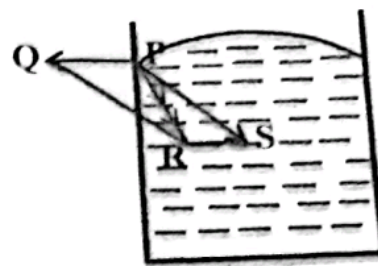


Figure3

The force actually acting on the molecule will be the resultant of these two forces of cohesion and adhesion. The force of cohesion acts along $\overrightarrow{PQ}$ at P. The force of adhesion acts at an angle of 45° to the vertical, and is represented by $\overrightarrow{PS}$. So that angle $QPS=135^\circ$. The resultant force is represented by parallelogram diagonal $\overrightarrow{PR}$. Its direction will depend on magnitude of $\overrightarrow{PQ}$ and $\overrightarrow{PS}$ and the following different cases arise.

- i) If $PQ/PS = 1/\sqrt{2}$, or $PS = \sqrt{2} PQ$, the resultant PR will lie along the vertically as shown in Fig.1.
- ii) If $PS < \sqrt{2} PQ$, then PR will outside the liquid as shown in Fig.2.
- iii) If $PS > \sqrt{2} PQ$, then PR will inside the liquid as shown in Fig.3.

Thus in the first case, when the cohesive force is $\sqrt{2}$ times the adhesive force, the molecules of the liquid are neither raised nor lowered and the liquid surface remains flat or plane.

In the second case, when the cohesive force is less than $\sqrt{2}$ times the adhesive force, the molecules of the liquid near the walls of the tube are raised up against the tube, making concave downward meniscus as in case of water.

In the third case when the cohesive force is greater than $\sqrt{2}$ times the adhesive force, the liquid meniscus near the walls of the tube depressed, making the surface convex upwards as in case of mercury.

*** Angle of contact:** When a liquid meets solid, its surface near its plane of contact with

solid is curved. The angle between the tangent to the liquid surface at the point of contact and solid surface inside the liquid is called the angle of contact for the pair of solids and liquids.

Capillarity and rise of liquid in a capillary Tube: A capillary tube is just a tube of very fine bore. One of the most striking effects of surface tension is to raise a liquid in a capillary tube dipped into it. It is the reason of surface tension and the event is called capillarity.

When a capillary tube is dipped in a liquid like water, which wets it, the liquid immediately raises up into it. If the bore of the tube be very fine, the shape of the liquid meniscus is spherical and concave upwards θ is very small.

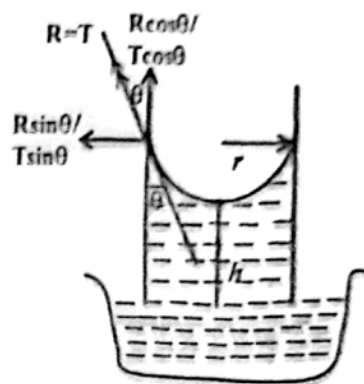
Let ' r ' be the radius of the tube so that the excess pressure on the meniscus is $\frac{2T}{r}$. If the liquid raise to a height ' h ' the hydrostatic pressure due to the liquid column in the tube will be $h\rho g$, where ' ρ ' is the density of the liquid. For equilibrium $\frac{2T}{r} = h\rho g$, or $T = \frac{r h \rho g}{2}$.

By knowing the r , h , ρ , and g the surface tension T of the liquid can be determined.

In case of tube containing narrow bore, the angle of contact, θ for the liquid is not zero and it has a finite value. At the contact point, surface tension force acts upward at angle θ with the vertical tube wall. The hydrostatic force acts just its opposite. In accordance with Newton's third law of motion, action and reaction are equal and opposite. This forces may be resolved two rectangular components, $R\cos\theta = T\cos\theta$ per cm along vertical line upward and $R\sin\theta = T\sin\theta$ along horizontal outward. If ' r ' be the radius of tube, the total upward force on the liquid in the tube is $2\pi r T \cos\theta$. Total volume of the liquid raised in the tube is $\pi r^2 h + V$ (V is volume of the meniscus water;

$$V = \frac{1}{3}\pi r^3). \quad \therefore 2\pi r T \cos\theta = (\pi r^2 h + V)\rho g = (\pi r^2 h + \frac{1}{3}\pi r^3)\rho g$$

$$\text{or } T = \frac{\pi r^2 (h + \frac{1}{3}r)\rho g}{2\pi r \cos\theta} \quad \text{In case of } \theta \approx 0^\circ, \cos\theta = 1 \text{ then } T = \frac{r(h + \frac{1}{3}r)\rho g}{2}$$



$$T = \frac{\pi r^2 \rho g (h + \frac{1}{3}r)}{2\pi \cos\theta}$$

Chapter 4: Elasticity

Elasticity: When a compression or extension force is applied causing a change in the shape or form of the body without significant change in its volume is removed, the body tends to recover its original condition. This property of a material body to regain its original condition, on the removal of deforming force is called elasticity.

Load, Stress and Strain: The term load implies the combination of external forces acting on a body and its effect is to change form or the dimensions of the body. It is also called deforming force.

Stress: When a load applied to a body, forces of reaction come into play internally in it tending to balance the load and restore the to its original condition. The restoring or recovering force per unit area setup inside the body is called stress.

Strain: The change produced in the dimensions of a body under a system of forces is called strain. The change in unit dimension is strain. (Linear, Volume and Shear strain.)

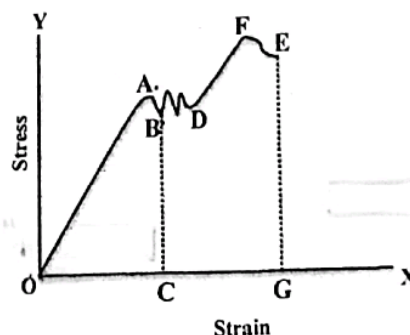
Hooke's law: (Within elastic limit, the stress is proportional to strain) so that the ratio stress/strain is a constant. This constant is called modulus of elasticity or coefficient of elasticity. When the stress is continuously increased in the case of solid, a point reached at which the strain increases more rapidly than stress. [The stress at which the linear relationship between stress and strain is invalid is known as elastic limit of material.]

Stress-strain curve: The relationship between stress and strain may be studied by plotting a graph between various values of stress and strain. This is called stress-strain diagram.

OA=Elastic limit, B=Yield point

OA=Elastic range, DE=Plastic range.

[See the Description from Elements of properties of matter, D. S. Mathur].



Corresponding to three different types of strain, there are three types of Elastic modulus or elastic constant.

- i) Linear elastic modulus or Young modulus corresponding to linear strain.
- ii) Volume elasticity or bulk modulus corresponding to volume strain and
- iii) Shear modulus or modulus of Rigidity corresponding to shear strain.

Young's modulus: When the deforming force is applied to the body along a perpendicular direction, the ratio of longitudinal stress to linear strain, within elastic limit is called Young's modulus and denoted by Y. *Young Modulus*, $Y = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}} = \frac{F/A}{l/L} = \frac{F.L}{A.l}$.

If $L=1$, $A=1$ and $l=1$, we have $Y=F$.

Hence, if a material of unit length and cross-section could be compressed/extended so as to increase/decrease in length by unity, the force is called Young's Modulus.

Bulk modulus: If a force is applied normally and uniformly to the whole surface of the body, so that there is a change of volume and no change of shape, the applied force per unit area gives the

stress and change per unit volume is volume strain. The ratio of these stress and strain is called Bulk modulus. The It is denoted by K .

$$\text{Bulk Modulus, } K = \frac{\text{Volume stress}}{\text{Volume strain}} = \frac{F/A}{v/V} = \frac{F.L}{A.v} = \frac{P.V}{v}$$

If $V=1$, and $v=1$, we have $K=F$.

Hence, if a material of unit volume could be compressed/extend so as to decrease/increase in volume by unity, the force is called Bulk Modulus.

Modulus of rigidity: The ratio of tangential stress to the shear strain gives the coefficient of the rigidity or modulus rigidity of the material of the body and denoted by η .

$$\text{Tangential stress} = \frac{F}{A}, \text{ shear strain} = \theta = \frac{l}{L}$$

$$\text{Modulus of rigidity, } \eta = \frac{F/A}{\theta} = \frac{F/A}{l/L} = \frac{F.L}{A.l}$$

If $A=1$, and $\theta=1$ radian, we have $\eta = F$.

Hence we may define modulus of rigidity of material as the shear stress or force per unit shear (1rad) ✓

Determination of Bulk modulus: Let ABCDEFGH be a unit cube and let forces T_x , T_y and T_z act perpendicularly to the faces BEHD & AFGC; ABDC & EFGH; and ABEF & DHGC respectively as shown in Figure 1. Then, if α be the

increment per unit length per unit tension along the direction of the force and β , be the contraction produced per unit tension in a direction perpendicular to the force, the elongation produced in the edges AB, BE and BD will obviously be $T_x\alpha$, $T_y\alpha$ and $T_z\alpha$ respectively and contraction

produced perpendicular to them will be $T_x\beta$, $T_y\beta$ and $T_z\beta$. The length of the edges thus become as follows: $AB = 1 + T_x\alpha - T_y\beta - T_z\beta$, $BE = 1 + T_y\alpha - T_x\beta - T_z\beta$, $BD = 1 + T_z\alpha - T_x\beta - T_y\beta$.

Hence the volume of the cube now becomes $(1 + T_x\alpha - T_y\beta - T_z\beta)(1 + T_y\alpha - T_x\beta - T_z\beta)(1 + T_z\alpha - T_x\beta - T_y\beta) = 1 + (\alpha - 2\beta)(T_x + T_y + T_z)$. Neglecting square and products of α and β , which are small compared with the other quantities involved. If $T_x = T_y = T_z = T$, the volume the cube becomes $1 + (\alpha - 2\beta).3T$. Therefore the increase in the volume of the cube

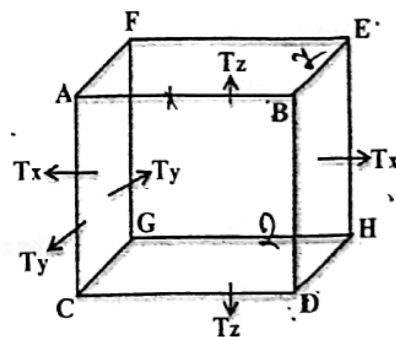
$$= 1 + (\alpha - 2\beta).3T - 1 = (\alpha - 2\beta).3T.$$

$$\text{The volume strain} = \frac{3T(\alpha - 2\beta)}{1}. \text{ Hence, Bulk modulus, } k = \frac{\text{stress}}{\text{strain}} = \frac{T}{3T(\alpha - 2\beta)}$$

[Stress = Tension per unit cross-section i.e $P=T/A$, for $A=1$, $P=T$]

$$\text{or } k = \frac{T}{3T(\alpha - 2\beta)}. \text{ The compressibility is reciprocal of Bulk modulus and becomes } C = 3(\alpha - 2\beta)$$

Determination of the Modulus of Rigidity: Let the top face ABHG of a cube be sheared by a shearing by a shearing force F , such that A goes to A' and B goes to B' and shearing angle $BCB' = \theta$. Tensile stress $= F/L^2 = T$ (say). Where L is the length of the edge of the cube.



Let the displacement $AA' = BB' = l$. Then shear strain $l/L = \theta$.
Coefficient of rigidity modulus, $\eta = T/\theta$.

If α and β be the longitudinal and the lateral strain per stress respectively, we have extension of diagonal DB due to tensile stress along it $= DB.T.\alpha$ and its extension due to compressive stress along AC $= DB.T.\beta$.

Therefore the total extension in the length of diagonal DB
 $= DB(T.\alpha + T.\beta) = DB.T(\alpha + \beta)$
 $= L\sqrt{2}.T(\alpha + \beta)$

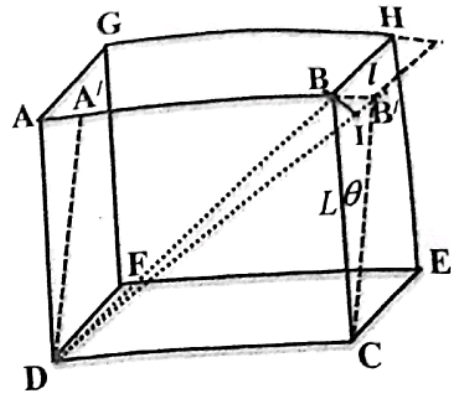
where $DB = \sqrt{L^2 + L^2} = L\sqrt{2}$

Let us draw a perpendicular BI from B onto DB'. Then

increase of length DB equal to IB'. Clearly $IB' = BB' \cos BB'I = l \cos 45^\circ = \frac{l}{\sqrt{2}}$.

Therefore $L\sqrt{2}.T(\alpha + \beta) = l/\sqrt{2}$ or $\frac{L.T}{l} = \frac{1}{2(\alpha + \beta)}$ or $\frac{T}{l/L} = \frac{1}{2(\alpha + \beta)}$ $\therefore \frac{T}{\theta} = \frac{1}{2(\alpha + \beta)}$

$\eta = \frac{1}{2(\alpha + \beta)}$ coefficient of rigidity or rigidity modulus.



Relation among the elastic constants:

We have the relation $k = \frac{1}{3(\alpha - 2\beta)}$ or $\alpha - 2\beta = \frac{1}{3k}$ (1)

and $\eta = \frac{1}{2(\alpha + \beta)}$ or $\alpha + \beta = \frac{1}{2\eta}$ (2)

Subtracting (1) from (2) we get $3\beta = \frac{1}{2\eta} - \frac{1}{3k} = \frac{3k - 2\eta}{6\eta k}$ or $\beta = \frac{3k - 2\eta}{18\eta k}$ (3)

Again multiplying (2) by 2 and adding with (1) we get $3\alpha = \frac{1}{\eta} + \frac{1}{3k} = \frac{3k + \eta}{3k\eta}$ or $\alpha = \frac{3k + \eta}{9k\eta}$

or $\frac{1}{Y} = \frac{3k + \eta}{9\eta k}$ [where young modulus, $Y = \frac{1}{\alpha}$]

or $Y = \frac{9k\eta}{3k + \eta}$ or $\frac{9}{Y} = \frac{3k + \eta}{k\eta} = \frac{3k}{k\eta} + \frac{\eta}{k\eta}$ or $\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{k}$ This is the relation between three elastic constants.

Poisson's ratio: Within the elastic limit, the lateral strain (β) is proportional to the linear or longitudinal strain (α) for the material of a given body and the ratio between two strain is a constant, called poisson's ratio for that material. It is denoted by σ . In engineering case $\sigma = \frac{\beta}{\alpha}$

Thus **Poisson's ratio** $= \frac{\text{lateral strain}}{\text{linear strain}} = \sigma = \frac{\beta}{\alpha} = \frac{1}{m}$

Relations for k and η interms of σ : We have the relation $k = \frac{1}{3(\alpha - 2\beta)}$

or $k = \frac{1}{3\alpha(1 - \frac{2\beta}{\alpha})} = \frac{1}{3\alpha(1 - 2\sigma)} = \frac{Y}{3(1 - 2\sigma)}$ where $\frac{1}{\alpha} = Y$

or $Y = 3k(1 - 2\sigma)$ or $Y = 3k(1 - \frac{2}{m})$ (1)

Again we have the relation $\eta = \frac{1}{2(\alpha + \beta)} = \frac{1}{2\alpha(1 + \frac{\beta}{\alpha})} = \frac{1}{2\alpha(1 + \sigma)}$ or $\eta = \frac{Y}{2(1 + \sigma)}$

Hence $Y = 2\eta(1 + \sigma)$ or $Y = 2\eta(1 + \frac{1}{m})$ (2)

Now from relation (1) and (2) $3k(1 - 2\sigma) = 2\eta(1 + \sigma)$ or $\sigma = \frac{1}{m} = \frac{3k - 2\eta}{6k + 2\eta}$

which gives the value of σ in terms of k and η .

Similarly if we eliminate σ from (1) and (2), we have $Y = \frac{9k\eta}{3k+\eta}$ or $\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{k}$.

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Limiting values of σ : We have the relation $3k(1 - 2\sigma) = 2\eta(1 + \sigma)$ where k and η are essentially positive quantities. Therefore, (i) if σ be a positive quantity, the left hand side and right hand side must be positive quantity. In that case $2\sigma < 1$ or $\sigma < 0.5$

and ii) if σ be a negative quantity, the left hand side and right hand side must be positive quantity. In that case σ be not less than -1. Hence we can write $-1 < \sigma < 0.5$

Cantilever: A cantilever is a beam fixed horizontally at one end and loaded at other end.

Bending of the Cantilever and expression of depression: Let AB represent the neutral axis of a cantilever of length L , fixed at A end and loaded at B end with weight, such that the end B is deflected or depressed into position B' and the neutral axis takes up the position AB' . Consider a point P of the beam at a distance x from the fixed end A. The bending moment of AP section due to load W on it = $W \cdot PB/2 = W(L-x)/2$.

Since the beam is in equilibrium, this must be equal to $Y I_g / R = Y \cdot a k^2 / R$, where R is the radius of curvature of the neutral axis at P. Bending moment of the beam = $(Y/R) \cdot I_g$.

Where the quantity $Y \cdot I_g = Y \cdot a k^2$ is called the flexural rigidity of the beam.

$$\text{Therefore } W(L-x) = \frac{Y \cdot I_g}{R} = \frac{Y \cdot a \cdot k^2}{R} \dots (1)$$

For a point Q at a small distance dx from P, the moment is almost same.

$$PQ = R \cdot d\theta \quad \text{or} \quad dx = R \cdot d\theta \quad \text{or} \quad R = dx/d\theta.$$

Substituting the value of R into (1), we have

$$W(L-x) = \frac{Y \cdot a \cdot k^2 \cdot d\theta}{dx} \quad \text{or} \quad d\theta = \frac{W \cdot (L-x) \cdot dx}{Y \cdot a \cdot k^2} \dots (2)$$

Let us draw the tangents to the neutral axis at P and Q, meeting the vertical line BB' at C and D. Then angle $C'PD = d\theta$. The depression of Q below P is equal to $CD = dy$ (say).

$$\text{Then } dy = (L-x) \cdot d\theta. \text{ Substituting for } d\theta, dy = \frac{(L-x) \cdot W \cdot (L-x) \cdot dx}{Y \cdot a \cdot k^2} = \frac{W \cdot (L-x)^2 \cdot dx}{Y \cdot a \cdot k^2} \dots (3)$$

The depression $y = BB'$ of the loaded end B is obtained by integrating the expression within limit $x=0$ and $x=L$.

$$\therefore y = \int_0^L \frac{W \cdot (L-x)^2}{Y \cdot a \cdot k^2} \cdot dx = \int_0^L (L^2 - 2xL + x^2) \cdot dx = \frac{W}{Y \cdot a \cdot k^2} \left[L^2 x - \frac{2x^2 L}{2} + \frac{x^3}{3} \right]_0^L = \frac{W}{Y \cdot a \cdot k^2} \left[L^3 - L^3 + \frac{L^3}{3} \right]$$

$$\text{Depression, } y = \frac{W \cdot L^3}{3 \cdot Y \cdot a \cdot k^2} = \frac{W \cdot L^3}{3 \cdot Y \cdot I_g}$$

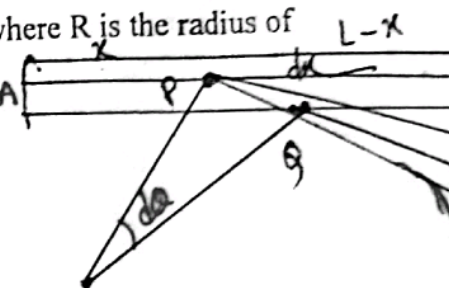


Figure: Bending beam

